Literature Comparison Table

Table 1: Paper Identification & Context

No	Title	Authors	Year/Venue	Dataset
1	Multiple Disease Prediction from Retinal Images Using Deep Learning Algorithm	Rishikesh, Karthikayani	2025, ICPCT	Kaggle (Cataract, DR, Glaucoma, Normal)
2	Multi-resolution and Multi-modal Feature Integration using GNNs for OCT Classification	Dissanayake et al.	2025, EMBC	OCTDL, OCTID, OCT-2014
3	Enhanced Multi-Scale Graph Networks for Glaucoma Detection in OCT	Mani et al.	2024, ICCCNT	Private OCT (71 pts), Duke SD-OCT
4	Explainable Graph-Based Spatial Vulnerability Mapping for Glaucoma Using Fundus Images	Ravi Kishan et al.	2026, ICSCAN	REFUGE + ORIGA (1,050 images)
5	Comprehensive Analysis of Automated Glaucoma Screening using Retinal Imaging and Explainable DNNs	Chandru, Vishagar	2025, IC3IT	Review (RMI-ONE, ACRIMA, REFUGE, ORIGA, RFMiD, etc.)

Table 2: Method & Performance

No	Application	Method	Metrics
1	Multi-disease fundus classification	VGG16 CNN, transfer learning	Acc 95–98.7%, AUC 0.96
2	Multi-class OCT classification	ResNet50+InceptionV3+ViT, GCN-based fusion (TTG+EFE)	Acc 94.5–99.8%, AUC up to 1.00
3	Optic disc + layer segmentation	U-Net + GCN graph reasoning, multi-scale	Dice 0.850, PA 0.895
4	Glaucoma detection + explainability	Dual CNN + Grad-CAM, RAG + Dijkstra routing	Acc 0.86, AUC 0.93, 61.8% ISNT alignment
5	Review of ML/DL glaucoma screening	Survey of CNN, GCN, GAN, DCNN, DnCNN, XAI	Varies by cited study

Table 3: Critical Appraisal

NO	Strengths	Limitations	Research Gap	Relation to Your Work
1	Simple, strong AUC, multi-disease	Low cataract precision, no clinical validation	No GNN implemented despite discussion	Non-graph CNN baseline
2	Strong SOTA margins, thorough ablations	Heavy compute, image-only fusion	Multimodal (clinical/text) fusion unexplored	Closest comparator; direct SOTA benchmark
3	Anatomical priors embedded, reduces artifacts	Small dataset, weaker on disc/RPE	Needs broader validation, disease scope	Segmentation-side GCN benchmark
4	Interpretable, clinically validated	Non-trainable graph, single-visit only	No learned-GNN + explainability fusion yet	Explainability-focused contrast case
5	Broad, current taxonomy	No original experiments	Data scarcity, compute cost, generalization	Context/positioning source

Table 4 (Track 1): GNN Type Detail

No	Learned GNN?	GNN Type	Graph Construction	Edge Weighting
1	No	—	Not implemented	—
2	Yes	GCN + graph attention (dynamic adjacency)	Feature embeddings from CNN/ViT layers, k-NN adjacency	Learned via VCR/VVR cross-attention
3	Yes	GCN (graph reasoning block)	Reshaped conv. feature maps, per pooling scale	Implicit in GCN message passing
4	No	Non-learned RAG + Dijkstra	Superpixels (SLIC, n=500)	Fixed: resistance = max(1−risk, 0.05)
5	Discussed only	GCN (general)	Not specified	Not specified